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BRIEF REVIEW

Analysis of Recovery Methods' Efficacy Applied up to 72 Hours Postmatch in Professional Football: A Systematic Review With Graded Recommendations

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Background: Sleep, nutrition, active recovery, cold-water immersion, and massage were recently reported as the most used postmatch recovery methods in professional football. However, the recommendations concerning the effect of these methods remain unclear. ***Purpose:*** To systematically review the literature regarding the effectiveness of the most common recovery methods applied to male and female football players (or other team sports) 72 hours postmatches and to provide graded recommendations for their use. ***Methods:*** A systematic search of the literature was performed, and the level of evidence of randomized and nonrandomized studies was classified as 1 or 2, respectively, with additional ++, +, and − classification according to the quality of the study and risk of bias. Graded recommendations were provided regarding the effectiveness of recovery methods for physical, physiological, and perceptive variables. ***Results:*** From the 3472 articles identified, 39 met the inclusion criteria for analysis. The studies' levels of evidence varied among methods (sleep: 2+ to 1++; nutrition: 2− to 1+; cold-water immersion: 2− to 1++; active recovery: 2− to 1+; and massage: 1− to 1+). Different graded recommendations were attributed, and none of them favored the effective use of recovery methods for physiological and physical parameters, whereas massage and cold-water immersion were recommended as beneficial for perceptive variables. ***Conclusions:*** Cold-water immersion and massage can be recommended to recover up to 72 hours postmatch at a perceptive level. However, there is a current need for high-quality research that identifies effective recovery strategies that enhance recovery at the physical and physiological levels.

Keywords: fatigue, elite football, soccer, recovery strategies, athletes

In professional football, the physiological demands of training and match play and the number of competitive matches played per season have increased in recent years.[1,2] Football players may play up to 60 competitive matches per season, representing 5 to 6 matches monthly and 3 matches per week during specific congested periods.[1,3] Notably, external workloads (eg, high-speed distance, sprint distance, accelerations, and decelerations) and the technical–tactical requirements (eg, number of passes, crosses, and shots on target) of match play have also increased.[4,5] Therefore, optimal recovery strategies are fundamental to avoid long-term fatigue and adverse consequences such as underperformance or injury.[6,7]

This trend has promoted the interest and the need to enhance the knowledge about effective postmatch recovery usage in professional football.[6,8–10] Recent research findings have revealed that the top 5 most used postmatch (up to 72 h) recovery methods in Portuguese professional football teams were sleep, nutrition, active recovery, cold water immersion, and massage.[10] However, concerning the effectiveness of these methods in recovery, it is still not clear whether they are beneficial and should be recommended to recover the players within 72 hours postmatch (ie, minimal time often considered to intersperse matches). The high controversy observed in studies has made it difficult to obtain clear conclusions and practical guidelines.[6,7,11–16] This may be justified by the limited number of randomized control trials performed with professional football players and the high heterogeneity observed within studies due to the divergency in participants' characteristics (ie, athletes, youth athletes, and active nonathletes), in the protocols of recovery and in the protocols used to induce fatigue and muscle damage.[6,11,12,17–21] In addition, a moderate to low methodological quality has been attributed to some of these studies',[7,12] which increases the risk of bias. However, the effectiveness of recovery methods can be determined through a systematic review design approach with clearer inclusion criteria and with graded recommendations attributed based on the studies' methodological quality. With this information, objective and evidence-based guidelines could be developed to guide recovery procedures in professional football.

Previous reviews related to recovery topics in football[6,7,11–16] missed important aspects such as: (1) some of the most popular postmatch recovery methods were not covered, (2) the evaluation of studies' methodological quality and the level of evidence were not performed, (3) graded recommendation was not assigned, and/or (4) critical methodological issues were not systematically accomplished (which complicates the replication of the study and the comparison of the data reported). Another important aspect that previous systematic reviews have also missed is to consider recovery in different domains (ie, physical, physiological, and perceptive).[6,7,11–16]

Therefore, a new and comprehensive systematic review that considers the effectiveness and graded recommendations of recovery methods in different domains postmatch (ie, physical,

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physiological, and perceptive) is warranted. In order to clarify the divergence between theory (ie, the effectiveness of recovery methods) and practice (ie, effective use of recovery methods), the current study aimed to systematically review the literature regarding the effectiveness of the commonly used recovery methods postmatch in professional football (ie, sleep, nutrition, active recovery, cold water immersion, and massage)[10] and to provide graded recommendations for their use based on the methodological quality of the studies, in either male or female athletes.

Methods

Design

A systematic review focusing on the effectiveness of recovery methods used postmatch in football was performed following the recommendations of the "Preferred Reporting Items for Systematic Reviews and Meta-Analyses" (PRISMA) statement[22] and based on the minimum criteria established by the Cochrane Back Review Group.[23] Before the literature search and selection procedure, a review protocol based on PRISMA-P[24] was completed (Supplementary Material S1 [available online]) and registered at the international prospective register of systematic reviews PROSPERO (ID: 272844).

Literature Search and Selection Process

A systematic literature search regarding the effectiveness of recovery methods used postmatch in football was performed within the following databases: PubMed (MEDLINE), SPORTDiscus, Scopus, ISI Web of Science, and Cochrane Library. The databases were searched using Boolean operators (ie, AND, OR, and NOT) with the combination of the following medical subject headings and synonyms associated with critical concepts of fatigue, recovery, performance, and well-being: "athlete," "team-sport," "recovery," "sleep," "nutrition," "active recovery," "cold water immersion," and "massage." The search strategies adopted were customized to accommodate the characteristics of each database and can be consulted in the Supplementary Material S2 (available online). In addition, a manual search of the reference lists provided in the selected papers was also performed. These searches and the screening procedures were independently performed by 2 of the study authors (S.M.Q. and R.R.) between August 16 and September 6, 2021, with differences resolved by consensus. Additionally, 2 external research experts in fatigue, recovery, and performance were invited to reduce the risk of missing relevant articles. The eligibility of the studies was formulated according to the following PICOS criteria:

- **P**opulation: football or other team sports athletes;
- **I**ntervention: the most used recovery methods in professional football reported in a previous study (ie, sleep, nutrition, active recovery, cold water immersion, and massage)[10] applied up to 72 hours after a match or exercise-induced muscle damage protocol;
- **C**omparison: between recovery methods (ie, sleep, nutrition, active recovery, cold water immersion, and massage) and a control group;
- **O**utcomes: physical performance (eg, maximal strength), physiological (eg, indirect markers of muscle damage), and perceptual (eg, well-being) outcomes collected up to 72 hours were considered;
- **S**tudy design: randomized and nonrandomized clinical trials.

Studies were included if written in English and published in peer-reviewed scientific journals, and if: (1) the participants were either adult male and/or female team sports athletes (ie, ≥18 y); (2) the intervention was performed for recovery purposes after a sport competition or related exercise-induced fatigue; and (3) the participants were allocated to at least 2 groups (ie, 1 control and 1 experimental group). Although the published survey which precedes the current systematic review was conducted in the context of professional football,[10] a wider athletic population of team sports was included to maximize the availability of studies and the extrapolation of evidence. Studies were excluded if they contained (1) nonteam sport athletes; (2) youth athletes (<18 y old); (3) other recovery methods not listed before; (4) a recovery intervention performed before or during the competition or exercise-induced fatigue; (5) a recovery intervention performed later than 72 hours post exercise-induced fatigue; (6) a recovery intervention performed after exercise conducted in altitude or in the heat; and (7) control groups receiving carbohydrates, in the case of nutrition studies (to avoid analysis bias). Returned titles and abstracts were downloaded into Mendeley Desktop (version 1.19, Glyph & Cog) and screened for eligibility criteria. After the removal of cross-references and duplicates, full articles were then selected based on the criteria set out above. Data were independently extracted by 2 researchers (S.M.Q. and R.R.), where extraction divergences were resolved by consensus; and the included studies' general characteristics and findings were systematically summarized in text and tables. The paucity of data collected for some recovery methods (ie, sleep, nutrition, and massage) and the methodological diversity observed between studies for all recovery methods limited potential attempts to undertake a meta-analysis.

Methodological Quality and Level of Evidence

The included studies' methodological quality and level of evidence were classified based on a checklist (ie, Downs and Black instrument),[25–27] composed of 27 questions, distributed between 5 subscales (ie, reporting, external validity, internal validity—bias, internal validity—confounding, and power). Two reviewers (S.M.Q. and R.R.) independently examined the studies quality as a percentage score,[25] with disagreements resolved by consensus. The level of evidence of randomized and nonrandomized studies was classified as 1 or 2, respectively, with an additional ++, +, and – classifications according to the quality of the study and risk of bias[25]: ++ for studies with high quality and very low risk of bias (≥75%); + for studies well conducted and with low risk of bias (50%–74%); and – for studies with low quality and high risk of bias (<50%).

Graded Recommendations

By following the guidelines of the Scottish Intercollegiate Guidelines Network (SIGN),[27] the grades of recommendation for physical performance, physiological, and perceptive outcomes of each recovery method were attributed by 4 sports science researchers qualified with a PhD during a round table. Based on the study's level of evidence, the population of the supported evidence (ie, from football or other team sports athletes), and the experts personal knowledge and experience, the recommendations either for or against each (sub)topic were categorized as described elsewhere[25,27]: A—strong recommendation; B—moderate recommendation; C—weak recommendation; and D—insufficient evidence to make a specific recommendation. Note that the type of recommendation for each recovery method (ie, to use or not to use) was provided depending on the studies' findings.

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Results

After the initial search (Figure 1), 3472 articles were identified, 250 duplicates were removed, 3222 titles and abstracts were screened, 3165 articles were excluded, and 8 records were added after the manual search of the reference lists provided in the selected papers. From the eligibility assessment (n = 65), 26 articles were excluded from qualitative synthesis and for methodological quality assessment due to the participants not being teams-sports athletes (n = 13), older than 18 years old (n = 12), and unavailability of 1 full article. The included studies (n = 39) evaluated the recovery postexercise of sleep (n = 3),[28–30] nutrition (n = 3),[31–33] cold water immersion (n = 21),[34–54] active recovery (n = 12),[42,43,55–64] and massage (n = 3).[51,65,66] The most relevant characteristics and findings are summarized in Supplementary Material S3 (available online).

Sleep

Two randomized[29,30] and 1 nonrandomized controlled trial[28] provided data related to sleep methods (Table 1). Two studies were conducted with amateur football players[28,30] and one with professional rugby players.[29] The studies varied in terms of sleep interventions after the match, namely when reporting the effect of napping in the afternoon,[28] sleeping in a high heat transfer capacity mattress during the night,[29] and using sleep hygiene strategies.[30] The study's methodological quality score ranged from 50.0% to 78.6%, whereas the levels of evidence ranged from 2+ to 1++ for physical performance and perceptive parameters and from 1+ to 1++ for physiological parameters. Given the findings from the studies reviewed for sleep strategies as a method of recovery in football, a strong recommendation (graded A) against the utilization of this method was assigned for all outcomes.

Nutrition

Two randomized[31,33] and 1 nonrandomized controlled trial[32] provided data related to the nutrition method (Table 2). Two studies were performed with professional football athletes[31,33] and one with amateur football athletes.[32] The nutrition strategy used after the match varied among studies, namely 30 g of whey protein,[31] semiskimmed milk,[32] and whey protein and high content of carbohydrates.[33] The study's methodological quality scores were 46.4% for physical performance and perceptive indicators and ranged from 35.7% to 56.1% for physiological parameters. Consequently, corresponding levels of evidence were 2− for physical performance and perceptive indicators, and ranged from 1− to 1+ for physiological parameters. Insufficient evidence to make a specific recommendation (graded D) was assigned to nutrition in all outcomes, whereas there is a lack of evidence to determine whether specific nutrition strategies influence recovery in physical, physiological, and perceptive parameters postcompetition.

Cold Water Immersion

Seventeen randomized[34–37,39–47,49–52] and 4 nonrandomized controlled trials[38,48,53,54] provided data related to cold water immersion (Table 3). The type of participants varied between studies, involving both amateur athletes of basketball (n = 1),[34,38] rugby (n = 4),[39,44,47,48] football (n = 1),[38] and other team sports (n = 4),[36,41,53,54] and professional athletes of basketball (n = 1),[51] rugby (n = 5),[35,37,49,50,52] football (n = 3),[40,42,43] futsal (n = 1),[45] and mixing team sports (n = 1).[46] The type of intervention varied among studies in terms of temperature, duration, and immersion method: (1) water temperature ranged between 5 and 10 °C in 13 studies[35–37,39–44,49,50,52,54] and between 11 and 15 °C in 8 studies,[34,41,45–48,51,53] (2) the duration of water immersion ranged between 1 and 10 minutes in 12 studies[35–40,44,47,49–51,54] and between 11 and 15 minutes in 9 studies,[34,41–43,45,46,48,52,53] and (3) the immersion method was continuous in 15 studies[34,36–48,53] and intermittent (ie, 2 × 5 min) in 6 studies.[35,49–52,54] The study's methodological quality score ranged from 42.9% to 85.7% for physical performance, 28.5% to 60.7% for physiological, and 42.9% to 67.9% for perceptive parameters, as well as the levels of evidence ranged from 2+ to 1++ (physical), 2− to 1+ (physiological), and 2+ to 1+ (perceptive). Given the findings from the studies reviewed for cold water immersion, a moderate recommendation (graded B) for the utilization of this method was assigned for perceptive outcomes. With a different trend, a moderate recommendation (graded B) against the utilization of cold water immersion was assigned for physical performance and physiological outcomes.

Active Recovery

Ten randomized[42,43,55–61,64] and 2 nonrandomized controlled trials[62,63] provided data related to active recovery (Table 4). These studies were performed with both amateur athletes of handball (n = 1), rugby (n = 1)[64], futsal (n = 1),[61] Australian rules football (n = 1),[63,64] football (n = 1),[55,56] and mixing team sports (n = 1)[56] and professional athletes of rugby (n = 1)[62] and football (n = 5).[42,43,57,59,60] The type of active recovery varied among studies by performing in the water (n = 3)[61,63,64] or in the land, using cycling (n = 3),[55,56,62] cycling and strength training (n = 2),[59,60] walking or running (n = 2),[58,61] or using walking or running and stretching exercises (n = 3).[42,43,57] The study's methodological quality score ranged from 42.9% to 71.4% for physical performance, 32.1% to 64.3% for physiological, and 42.9 to 64.3 for perceptive parameters. Consequently, the levels of evidence ranged from 2+ to 1+ (physical performance), 2− to 1+ (physiological), and 2+ to 1+ (perceptive). Given the findings from the studies reviewed for active recovery strategies as a method of recovery in football, a moderate recommendation (graded B) against the utilization of this method was assigned for all outcomes.

Massage

Three randomized controlled trials[51,65,66] were included (Table 5), where 2 studies were performed with professional basketball players[51,65] and one with amateur team sports players.[66] In all studies, massages were performed up to 30 minutes after a match or exercise with a similar approach. The study's methodological quality scores ranged from 46.4% to 64.3% for physical and perceptive parameters, with the levels of evidence ranging from 1− to 1+ in both outcomes. None of these studies assessed physiological parameters. Given the findings from the studies reviewed for massage strategies, a moderate recommendation (graded B) for the utilization of this method was assigned for perceptive outcomes. With a different tendency, a moderate recommendation (graded B) against the utilization of massage was assigned for physical performance outcomes, and nonexistent evidence was observed for physiological outcomes.

Discussion

Considering the importance of ensuring an effective recovery after football matches,[6–9] the present study aimed to systematically

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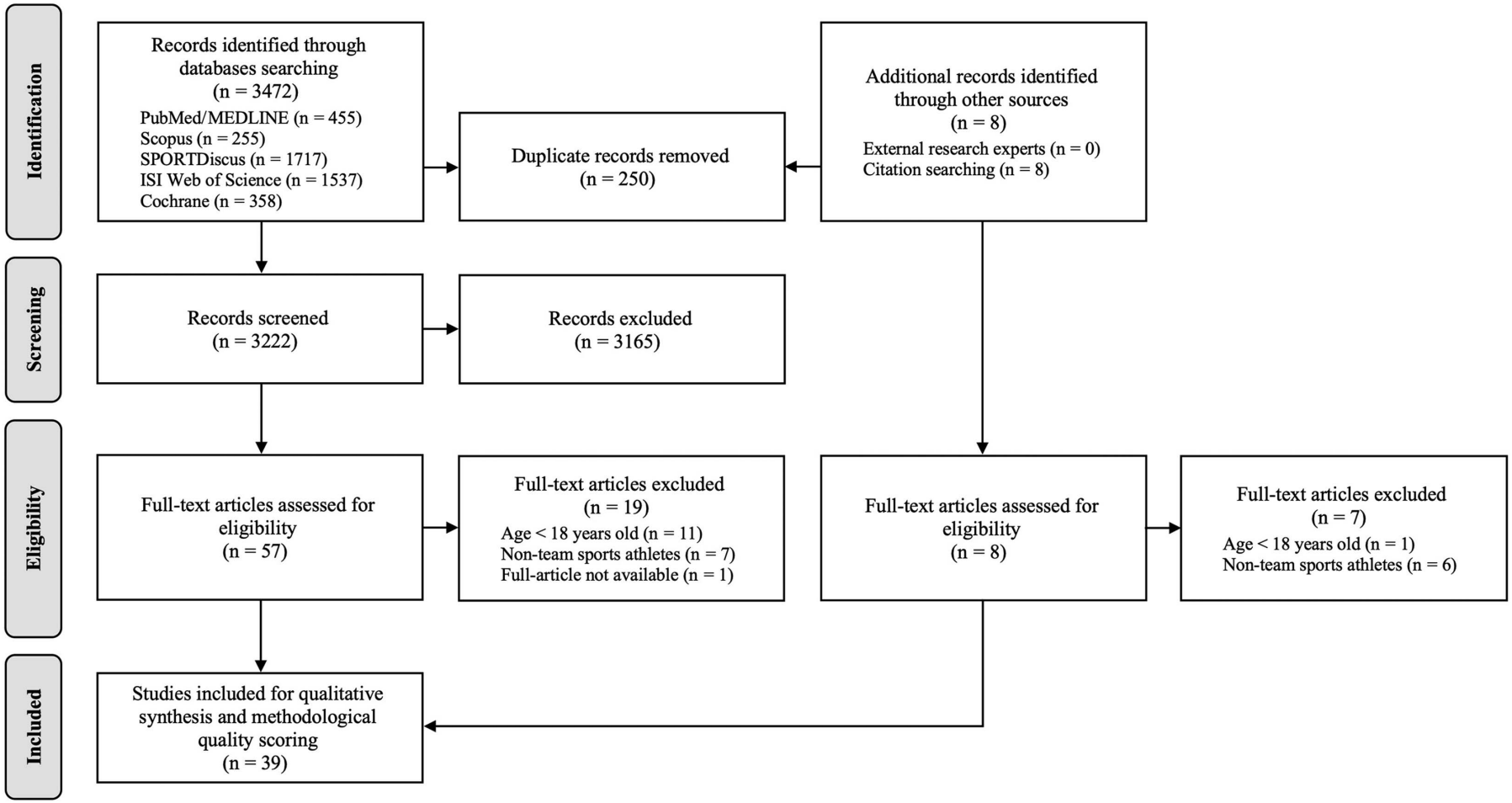

Figure 1 — Flowchart to illustrate the studies' identification, screening, eligibility, and inclusion.

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Table 1 Graded Recommendations Assigned to the Physical Performance, Physiological, and Perceptive Outcomes, as Well as the Quality Score, Scientific Level of Evidence, and the Main Findings for Articles Investigating the Effect of Sleep in Recovery

Study	Design	Participants	Quality score, %	Level of evidence	Main findings
Physical performance parameters (n = 3) Graded recommendation: graded A, sleep does not influence recovery of physical performance parameters					
Hsouna et al[28]	NRCT	Amateur football players	53.6	2+	↑ anaerobic performance (ie, 5-m shuttle) at ~20 h
Aloulou et al[29]	RCT	Elite rugby players	78.6	1++	= anaerobic performance (ie, vertical jump) at 24 h
Fullagar et al[30]	RCT	Amateur football players	50.0	1+	= anaerobic performance (ie, vertical jump) at 10, 20, 34, and 44 h; = aerobic performance (ie, YYIRT2) at 12 and 36 h
Physiological parameters (n = 2) Graded recommendation: graded A, sleep does not influence recovery of physiological markers					
Aloulou et al[29]	RCT	Elite rugby players	78.6	1++	= damage markers (ie, CK) at 24 h
Fullagar et al[30]	RCT	Amateur football players	50.0	1+	= damage (ie, CK and urea) and = inflammatory markers (ie, CPR) at 10, 20, 34, and 44 h
Perceptive parameters (n = 3) Graded recommendation: graded A, sleep does not influence perceptive recovery					
Hsouna et al[28]	NRCT	Amateur football players	53.6	2+	↑ muscle soreness at ~18 h; ↑ sleepiness at ~18 h
Aloulou et al[29]	RCT	Elite rugby players	78.6	1++	= general fatigue at 24 h
Fullagar et al[30]	RCT	Amateur football players	50.0	1+	= perceptive recovery and = stress at 12, 24, 36, and 48 h

Abbreviations: CK, creatine kinase; CPR, C-reactive protein; n, number of studies; NRCT, nonrandomized control trial; RCT, randomized control trial; YYIRT2, Yo-Yo Intermittent Recovery Test level 2.
Note: ↑ significant benefit, ↓ significant harm, and = no significant difference.

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Table 2 Graded Recommendations Assigned to the Physical Performance, Physiological, and Perceptive Outcomes, as Well as the Quality Score, Scientific Level of Evidence, and the Main Findings for Articles Investigating the Effect of Nutrition in Recovery

Study	Design	Participants	Quality score, %	Level of evidence	Main findings
Physical performance parameters (n = 1) Graded recommendation: graded D, nutrition does not influence recovery of physical performance parameters					
Cockburn et al[32]	NRCT	Amateur football players	46.4	2–	= anaerobic performance (ie, vertical jump, 15-m sprint and agility) at 24, 48, and 72 h
Physiological parameters (n = 3) Graded recommendation: graded D, nutrition does not influence recovery of physiological markers					
Mor and Ipekoglu[31]	RCT	Amateur football players	35.7	1–	Controversy in damage markers (ie, ↑ AST but = LDH and = CK) at 2 h; = protein metabolism marker (ie, total protein) at 2 h
Cockburn et al[32]	NRCT	Amateur football players	46.4	2–	= damage markers (ie, CK and CK-Mb) at 24, 48, and 72 h
Gunnarsson et al[33]	RCT	Elite football players	56.1	1+	↓ damage markers (ie, CK) at 24 h; = carbohydrates metabolism markers (ie, muscle glycogen and glycogen resynthesis) at 24 h
Perceptive parameters (n = 1) Graded recommendation: graded D, nutrition does not influence perceptive recovery					
Cockburn et al[32]	NRCT	Amateur football players	46.4	2–	= muscle soreness at 24, 48, and 72 h

Abbreviations: AST, aspartate transaminase; CK, creatine kinase; CK-Mb, creatine kinase myocardial; LDH, lactate dehydrogenase; n, number of studies; NRCT, nonrandomized control trial; RCT, randomized control trial. Note: ↑ significant benefit, ↓ significant harm, and = no significant difference.

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Table 3 Graded Recommendations Assigned to the Physical Performance, Physiological, and Perceptive Outcomes, as Well as the Quality Score, Scientific Level of Evidence, and the Main Findings for Articles Investigating the Effect of Cold Water Immersion in Recovery

Study	Design	Participants	Quality score, %	Level of evidence	Main findings
Physical performance parameters (n = 18) Graded recommendation: graded B, cold water immersion does not influence recovery of physical performance parameters					
Chaiyakul and Chaibal[34]	RCT	Amateur basketball players	53.4	1+	= aerobic performance at 24 h; ↑ anaerobic performance (ie, vertical jump) at 24 h
Barber et al[35]	RCT	Elite rugby players	53.4	1+	Likely benefits in anaerobic performance (ie, lower limb strength and vertical jump) at 24 and 48 h
Egaña et al[36]	RCT	Amateur team sports players	46.4	1−	= cycling performance (ie, mean power) at ~0 h
Nunes et al[37]	RCT	Elite rugby players	50.0	1+	Controversy in anaerobic performance (ie, likely benefits in SJ but trivial effects in CMJ and 30-m sprint) at 24 and 48 h
Chow et al[39]	RCT	Amateur rugby players	85.7	1++	= anaerobic performance (ie, lower limb strength, vertical jump and agility) at ~0 h
Bouzid et al[40]	RCT	Elite football players	50.0	1+	Controversy in anaerobic performance (ie, ↓ SJ but = CMJ and = sprint) at ~0 h; = anaerobic performance (ie, vertical jump, lower limb strength, and sprint) at 24, 48, and 72 h
Anderson et al[41]	RCT	Amateur team sports players	53.4	1+	↓ cycling performance (ie, mean power) at ~0, 24, 48, and 72 h
Garcia et al[44]	RCT	Amateur rugby players	67.9	1+	↓ anaerobic performance (ie, agility and vertical jump) at ~0; = anaerobic performance (ie, agility and vertical jump) at 24 h
Moreira et al[45]	RCT	Elite futsal players	50.0	1+	↓ anaerobic performance (ie, vertical jump and repeated sprint) at ~0; = anaerobic performance (ie, vertical jump and repeated sprint) at 24 h
Leeder et al[46]	RCT	Elite team sports players	60.7	1+	= anaerobic performance (ie, lower limb strength and vertical jump) at ~0, 24, 48, and 72 h
Takeda et al[47]	RCT	Amateur rugby players	50.0	1+	= anaerobic (ie, 50-m sprint, vertical jump, and reaction time) and = cycling performance at ~0 and 24 h.
Cook and Beaven[48]	NRCT	Amateur rugby players	57.1	2+	↑ anaerobic performance (ie, repeated sprints) at 24 h
Higgins et al[49]	RCT	Elite rugby players	42.9	1−	= flexibility and = anaerobic performance (ie, vertical jump) at ~0, 24, and 48 h
Higgins et al[50]	RCT	Elite rugby players	46.4	1−	= anaerobic performance (ie, vertical jump and 40-m sprint) at ~0, 24, 48, and 72 h
Delextrat et al[51]	RCT	Elite basketball players	64.3	1+	Controversy in anaerobic performance (ie, ↑ vertical jump but = repeated sprint) at 24 h
Pointon and Duffield[52]	RCT	Elite rugby players	57.1	1+	↑ anaerobic performance (ie, lower limb strength and voluntary activation) at ~0 h; = anaerobic performance (ie, lower limb strength and voluntary activation) at 24 h
Brophy-Williams et al[53]	NRCT	Amateur team sports players	50.0	2+	↑ aerobic performance (ie, YYIRT1) at 24 h
Ingram et al[54]	NRCT	Amateur team sports players	57.1	2+	Controversy in anaerobic performance (ie, ↑ repeated sprints but = lower limb strength) at 48 h

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Table 3 *(continued)*

Study	Design	Participants	Quality score, %	Level of evidence	Main findings
Physiological parameters (n = 13) Graded recommendation: graded B, cold water immersion does not influence recovery of physiological parameters					
Chaiyakul and Chaibal[34]	RCT	Amateur basketball players	53.4	1+	= hemodynamic markers (ie, HR and blood pressure) at 24 h
Barber et al[35]	RCT	Elite rugby players	53.4	1+	Likely benefits in damage markers (ie, CK) at 24 and 48 h
Nunes et al[37]	RCT	Elite rugby players	50.0	1+	Controversy in inflammatory markers (ie, likely benefits in TNF-α but trivial effect in Il-6) at 24, 48, and 72 h
Çakir et al[38]	NRCT	Amateur football players	39.3	2−	= damage markers (ie, CK-Mb and cTnT) at 2 and 24 h
Bouzid et al[40]	RCT	Elite football players	50.0	1+	↑ damage marker (ie, CK) at post and 24 h; = damage marker (ie, CK) at 48 and 72 h
Anderson et al[41]	RCT	Amateur team sports players	53.4	1+	= damage (ie, CK), = inflammatory (ie, CPR), and = carbohydrates metabolism marker (ie, lactate) at ~0, 24, 48, and 72 h
Abedi et al[42]	RCT	Elite football players	28.6	1−	= damage (ie, CK) and = inflammatory marker (ie, CPR) at ~0 h; ↑ damage (ie, CK) and ↑ inflammatory marker (ie, CPR) at 24 and 48 h
Arabmomeni and Mostafavi[43]	RCT	Elite football players	32.1	1−	↑ damage (ie, CK) and ↑ inflammatory marker (ie, CPR) at 24 and 48 h
Leeder et al[46]	RCT	Elite team sports players	60.7	1+	= damage (ie, CK) and = inflammatory marker (ie, CPR and Il-6) at ~0, 24, 48, and 72 h
Takeda et al[47]	RCT	Amateur rugby players	50.0	1+	= damage (ie, AST, LDH creatinine, and CK) and = carbohydrates metabolism marker (ie, lactate) at ~0 and 24 h
Pointon and Duffield[52]	RCT	Elite rugby players	57.1	1+	= damage (ie, AST and CK) and = inflammatory markers (ie, CPR) at ~0, 2, and 24 h
Brophy-Williams et al[53]	NRCT	Amateur team sports players	50.0	2+	↑ inflammatory markers (ie, CPR) at 24 h; = carbohydrates metabolism marker (ie, lactate) at 24 h; = hemodynamic markers (ie, HR) at 24 h
Ingram et al[54]	NRCT	Amateur team sports players	57.1	2+	= damage (ie, CK) and = inflammatory markers (ie, CPR) at ~0, 24, and 48 h

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Table 3 *(continued)*

Study	Design	Participants	Quality score, %	Level of evidence	Main findings
Perceptive parameters (n = 15) Graded recommendation: graded B, cold water immersion may positively influence the perceptive recovery					
Chaiyakul and Chaibal[34]	RCT	Amateur basketball players	53.4	1+	↑ muscle soreness at 24 h
Barber et al[35]	RCT	Elite rugby players	53.4	1+	Likely benefits in muscle soreness at 24 and 48 h
Nunes et al[37]	RCT	Elite rugby players	50.0	1+	Trivial effect in muscle soreness and general fatigue at 24, 48, and 72 h
Bouzid et al[40]	RCT	Elite football players	50.0	1+	↑ muscle soreness at ~0, 24, and 48 h; = muscle soreness at 72 h
Anderson et al[41]	RCT	Amateur team sports players	53.4	1+	= muscle soreness at ~0, 24, 48, and 72 h
Garcia et al[44]	RCT	Amateur rugby players	67.9	1+	= perceptive recovery at ~0; ↑ perceptive recovery at 12 h
Moreira et al[45]	RCT	Elite futsal players	50.0	1+	= muscle soreness at ~0 and 24 h
Leeder et al[46]	RCT	Elite team sports players	60.7	1+	= muscle soreness at ~0, 24, 48, and 72 h
Takeda et al[47]	RCT	Amateur rugby players	50.0	1+	↑ general fatigue at ~0; = general fatigue at 24 h
Cook and Beaven[48]	NRCT	Amateur rugby players	57.1	2+	↑ perceptive recovery at 24 h
Higgins et al[49]	RCT	Elite rugby players	42.9	1−	= muscle soreness at ~0, 24, and 48 h
Higgins et al[50]	RCT	Elite rugby players	46.4	1−	= muscle soreness at ~0, 24, 48, and 72 h
Delextrat et al[51]	RCT	Elite basketball players	64.3	1+	↑ general fatigue and ↑ muscle soreness at ~0 and 24 h
Pointon and Duffield[52]	RCT	Elite rugby players	57.1	1+	↑ muscle soreness at 2 h; = muscle soreness at ~0 and 24 h
Ingram et al[54]	NRCT	Amateur team sports players	57.1	2+	= muscle soreness at ~0 h; ↑ muscle soreness at 24 and 48 h

Abbreviations: AST, aspartate transaminase; CK, creatine kinase; CK-Mb, creatine kinase myocardial; CMJ, countermovement jump; CPR, Creactive protein; cTnT, cardiac troponin T; HR, heart rate; Il-6, interleukin-6; LDH, lactate dehydrogenase; n, number of studies; NRCT, nonrandomized control trial; RCT, randomized control trial; SJ, squat jump; TNF-α, tumor necrosis factor alpha; YYIRT1, Yo-Yo Intermittent Recovery Test level 1. Note: ↑ significant benefit, ↓ significant harm, and = no significant difference.

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Table 4 Graded Recommendations Assigned to the Physical Performance, Physiological, and Perceptive Outcomes, as Well as the Quality Score, Scientific Level of Evidence, and the Main Findings for Articles Investigating the Effect of Active Recovery in Recovery

Study	Design	Participants	Quality score, %	Level of evidence	Main findings
Physical performance parameters (n = 5) Graded recommendation: graded B, active recovery does not influence recovery of physical performance parameters					
Abaïdia et al[55]	RCT	Amateur basketball players	42.9	1−	Trivial effect in anaerobic performance (ie, vertical jump and lower limb strength) at 48 and 72 h; likely harm in anaerobic performance at 20 h (ie, lower limb strength)
Rey et al[57]	RCT	Amateur football players	71.4	1+	Controversy in anaerobic performance (ie, ↑ vertical jump but = 20-m sprint and = agility at 24 h); = flexibility at 24 h
Andersson et al[60]	RCT	Amateur team sports players	64.3	1+	= anaerobic performance (ie, sprint, vertical jump, and lower limb strength) at ~0, 5, 21, 27, 45, 51, 69, and 74 h
Tessitore et al[61]	RCT	Elite football players	60.7	1+	= anaerobic performance (ie, vertical jump and 10-m sprint) at ~5 and ~21 h
Dawson et al[63]	NRCT	Amateur rugby players	53.4	2+	↑ anaerobic performance (ie, vertical jump) at 15 h but = anaerobic performance (ie, vertical jump) at 48 h; = cycling performance and = flexibility at 15 and 48 h
Physiological parameters (n = 10) Graded recommendation: graded B, active recovery does not influence recovery of physiological parameters					
Abaïdia et al[55]	RCT	Amateur basketball players	42.9	1−	Trivial effect in damage marker (ie, CK) at 20, 48, and 72 h
Özsu et al[56]	RCT	Amateur rugby players	50.0	1+	↑ carbohydrates metabolism marker (ie, lactate) at ~0 h; = hemodynamic markers (ie, HR) at ~0 h
Abedi et al[42]	RCT	Elite football players	28.6	1−	= damage (ie, CK) and = inflammatory marker (ie, CPR) at ~0 and 24 h; = damage (ie, CK) and ↑ inflammatory marker (ie, CPR) at 48 h
Arabmomeni and Mostafavi[43]	RCT	Elite football players	32.1	1−	= damage (ie, CK) and = inflammatory marker (ie, CPR) at 24 and 48 h
Arazi et al[30]	RCT	Amateur handball players	32.1	1−	↑ carbohydrates metabolism marker (ie, lactate) at ~0 h; ↓ hemodynamic markers (ie, HR and blood pressure) at ~0 h
Andersson et al[59]	RCT	Elite football players	60.7	1+	= oxidative stress and = antioxidants markers at ~0, 21, 45, 69, and 74 h
Andersson et al[60]	RCT	Amateur team sports players	64.3	1+	= damage markers (ie, CK, urea, and uric acid) at ~0, 5, 21, 27, 45, 51, 69, and 74 h
Tessitore et al[61]	RCT	Elite football players	60.7	1+	= stress (ie, cortisol and catecholamines) and = carbohydrates metabolism marker (ie, lactate) at ~20 h
Gill et al[62]	NRCT	Elite football players	46.4	2−	= damage markers (ie, CK) at ~0 h; ↑ damage markers (ie, CK) at 36 and 84 h
Suzuki et al[64]	NRCT	Amateur team sports players	53.4	1+	= damage (ie, CK, LDH, GPT, and GOT) and inflammatory markers (ie, leukocyte and neutrophil) at 24 and 48 h

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Table 4 *(continued)*

Study	Design	Participants	Quality score, %	Level of evidence	Main findings
Perceptive parameters (n = 6) Graded recommendation: graded B, active recovery does not influence perceptive recovery					
Abaïdia et al[55]	RCT	Amateur basketball players	42.9	1−	Likely harm effect on perceptive recovery at 20, 48, and 72 h
Özsu et al[56]	RCT	Amateur rugby players	50.0	1+	↑ perceptive recovery at ~0 h
Andersson et al[60]	RCT	Amateur team sports players	64.3	1+	= muscle soreness at ~0, 5, 21, 27, 45, 51, 69, and 74 h
Tessitore et al[61]	RCT	Elite football players	60.7	1+	= muscle soreness, = stress, = perceptive recovery and = amount of sleep at ~20 h
Dawson et al[63]	NRCT	Amateur rugby players	53.4	2+	= muscle soreness at 15 and 48 h
Suzuki et al[64]	NRCT	Amateur team sports players	53.4	1+	= mood (ie, POMS) at 24 and 48 h

Abbreviations: CK, creatine kinase; CPR, Creactive protein; GOT, glutamic oxaloacetic transaminase; GPT, glutamic pyruvic transaminase; HR, heart rate; LDH, lactate dehydrogenase; POMS, profile of mood states; n, number of studies; NRCT, nonrandomized control trial; RCT, randomized control trial. Note: ↑ significant benefit, ↓ significant harm, and = no significant difference.

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Table 5 Graded Recommendations Assigned to the Physical Performance, Physiological, and Perceptive Outcomes, as Well as the Quality Score, Scientific Level of Evidence, and the Main Findings for Articles Investigating the Effect of Massage in Recovery

Study	Design	Participants	Quality score, %	Level of evidence	Main findings
Physical performance parameters (n = 3) Graded recommendation: graded B, massage does not influence recovery of physical performance parameters					
Delextrat et al[65]	RCT	Elite basketball players	46.4	1−	= anaerobic performance (ie, vertical jump and repeated sprints) at 24 h
Delextrat et al[51]	RCT	Elite basketball players	64.3	1+	= anaerobic performance (ie, vertical jump and repeated sprints) at 24 h
Mancinelli et al[66]	RCT	Amateur team sports players	64.3	1+	↑ anaerobic performance (ie, vertical jump) at ~0 h; = aerobic performance and = flexibility at ~0 h
Physiological parameters (n = 0) Graded recommendation: none					
Perceptive parameters (n = 3) Graded recommendation: graded B, massage may positively influence the perceptive recovery					
Delextrat et al[65]	RCT	Elite basketball players	46.4	1−	↑ muscle soreness and ↑ general fatigue at ~0 and 24 h
Delextrat et al[51]	RCT	Elite basketball players	64.3	1+	↑ muscle soreness and ↑ general fatigue at ~0 and 24 h
Mancinelli et al[66]	RCT	Amateur team sports players	64.3	1+	↑ muscle soreness at ~0 h

Abbreviation: n, number of studies; NRCT, nonrandomized control trial; RCT, randomized control trial. Note: ↑ significant benefit, ↓ significant harm, and = no significant difference.

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review whether the recovery methods most commonly used in professional football (ie, sleep, nutrition, active recovery, cold water immersion, and massage)[10] are supported by scientific evidence and to provide graded recommendations on the respective use and effectiveness. Despite that the level of evidence and the graded recommendations were moderate to strong in most of the recovery methods, it was generally concluded that cold water immersion and massage can provide benefits to recover perceptive parameters. However, cold water immersion and massage did not show benefits in recovery of other parameters (i.e., physical performance and physiological outcomes). On the other hand, due to the limited number of studies observed for nutrition, sleep, and massage, there is a lack of evidence to recommend or avoid their use to recover physical and physiological outcomes. These findings encourage a rethinking of the common and current recovery practices postmatch, as well as the need to perform further studies with appropriate experimental designs (ie, randomized control trials), higher methodological quality, and lower risk of bias.

Sleep

Although sleep has been referred to as one of the most important[11,14,67] and frequent[8,9] postmatch recovery methods, and that sleep quality and quantity are normally poor in professional team sports athletes,[11,68] we found that sleep strategies appear to have negligible effectiveness in potentiating recovery at the physical, physiological, and perceptual levels. It should be noted that among the 3 studies considered, only one promoted sleep extension,[28] which has been reported as the most effective and advantageous sleep strategy for the recovery process[67]; however, sleep extension has been promoted through an educational session. Also, previous studies suggesting the importance of sleep for recovery effectiveness have focused on how sleep deprivation affected recovery,[11,67,69] rather than how additional sleep strategies might enhance recovery. This suggests that sleep strategies should not be considered as a way to recover in the short term over the impact of match (or to reduce recovery time), but rather in the perspective of safeguarding normal sleep practices of athletes and recovery in the long term. For instance, Bonnar et al[67] noted that sleeping strategies promote benefits in recovery only after long-term interventions (ie, some weeks of implementation). However, it is important to note that sleep deprivation has been reported to affect cognitive performance, decision making, and motor control behavior,[14,67,69] and these aspects, which are very important in the context of football, have been little studied in terms of recovery. Currently, cognitive-related outcomes may be more favored by good sleep practices than other physical and physiological outcomes. Therefore, despite the strong recommendation (graded A) supporting that sleep strategies do not influence recovery (which is in line with previous studies),[12,67] it should be noted that this does not mean that regular sleep strategies should be disregarded. Thus, we recognize the need of future controlled trials examining the impact of long-term sleep strategies on recovery of team sports athletes, in particular in variables likely to be disturbed by sleep.

Nutrition

Contrary to the considerable importance and extensive usage in professional football,[8–10] the effectiveness of nutrition strategies used postmatch needs to be further examined. The findings are contrary to previous studies supporting that various nutritional approaches (ie, fluids ingestion, carbohydrates and protein, tart cherry, omega-3, vitamin D, and creatine monohydrate supplementation) are beneficial to recover physical performance (eg, maximal strength), physiological (eg, rehydration, muscle glycogen refueling, and repair), and perceptual (eg, muscle soreness) parameters.[6,13,16,70,71] However, most of the conclusions of these studies derived from narrative reviews and consensus statements.[6,13,70,71] Also, the recommendations are mostly based on the effects induced by harmful exercise and the presumed postexercise nutritional positive effects, rather than on well-designed controlled trials.[6,13,70] Some of the studies have conducted long-term interventions rather than short-term interventions postexercise up to 72 hours,[6,16] and, important to note, none of the studies included have examined team sports athletes after competitive matches. It is also important to highlight the limitations of the studies included in the current review. The low methodological quality (ie, 35.7%–56.1%) and the research designs of the studies led to grade nutritional strategies to the lowest level of recommendation (graded D; insufficient evidence to make a specific recommendation). Thus, to date, there is no evidence that isolated nutritional strategies applied postmatch produce a beneficial effect on recovery up to 72 hours,[31–33] which opposes the recommendation of Abaïdia and Dupont.[12] Additionally, the nutritional strategies chosen (ie, whey protein, milk, carbohydrates, and protein) in the included studies[31–33] may not effectively favor the recovery process in the short term. Therefore, future studies should consider robust experimental designs with high methodological quality (ie, randomized control trials) aiming to test and recommend with greater confidence the use of nutrition strategies in the recovery of football players.

Cold Water Immersion

Although cold water immersion has been widely used for recovery purposes,[10] this method proved to be effective only to recover at a perceptual level, which is in agreement with most studies on this topic.[19,21,72–74] On the contrary, cold water immersion was not effective in recovering physical performance and physiological parameters, which is contrary to the conclusions of some studies[21,72,75] but not others.[19,76,77] The different findings between studies may be explained by the high bias and the heterogeneity of studies reported in previous systematic reviews.[19,21,73,75,76] For instance, the characteristics of the participants (eg, age, sex, physique trait) and the type of intervention (eg, duration, water temperature) may influence the magnitude of recovery due to the different effects in blood flow and tissue temperature.[18,78] Thus, some new approaches focusing on the individualization of cold water immersion to enhance the benefits in recovery were recently proposed.[18,78]

According to previous reports,[78] cold water immersion may induce deleterious effects in anaerobic performance at 24 hours postexercise.[41,44,45] This negative effect may be explained by the reduction in the neural drive and enzymatic activity,[44,78] although the relevance of this phenomenon may have limited applicability in professional football. In addition, the long-term use of cold water immersion may be potentially harmful to skeletal muscle mass development, in particular when using more aggressive protocols with lower temperatures and higher duration.[78,79] Although there are few studies concerning the long-term implications of using cold water immersion for recovery, none has been conducted in team sports, and it might be wise to use cold water immersion as per the recommendations (ie, water temperature at 11–15 °C and duration of 11–15 min),[73,75] and periodically in strictly necessary moments (eg, congested fixtures or after highly harmful exercise). Although the practical suggestions noted to avoid potential harmful effects,

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the overall high methodological quality of the studies and level of evidence reported reinforce the beneficial use of cold water immersion after football matches, in particular aiming to accelerate perceptive recovery, which is in line with Abaïdia and Dupont[12] and Cullen et al.[15]

Active Recovery

In team sports, active recovery may be performed immediately postexercise (ie, as a cooldown)[8,68] or in later periods up to 72 hours.[8–10] Particularly in football, active recovery has been mostly used between 24 and 72 hours postmatch.[10] This recovery strategy has been reported to be well accepted by the practitioners in charge and positively assumed by athletes as effective for recovery.[6,8–10] However, the effects of exercise in the recovery of physical, physiological, and perceptive outcomes have been reported as not significant,[6,68] which is in agreement with the findings of the present study. It is important to note, however, that postmatch active recovery practices usually incorporate socialization actions and promotion of squad motivation among players (usually valued in postgame "psychological recovery") and may allow coach interventions on technical–tactical aspects. As such, although a moderate recommendation against the use of active recovery (ie, grade B) is given, such strategies might not be disregarded. Future research on active recovery should seek to find out whether such practices effectively produce psychosocial benefits and whether self-selected active recovery strategies may promote better recovery at the individual level.

Massage

Although with considerably less degree of usage for football postmatch recovery compared with the aforementioned methods,[10] previous research has reported controversial findings for massage in recovery of physical and physiological outcomes, with some systematic reviews showing benefits[21,80] and others not.[6,12,20,81,82] It has been suggested that this controversy may be explained by the large variety of massage techniques used among the analyzed studies.[6] In the present work, the 3 included studies used similar techniques and protocols and showed different findings on the use of massage for physical performance recovery; 1 study [66] reported benefits and 2 studies [51,65] reported no advantages. On the other hand, all studies showed positive effects of massage for perceptive recovery,[51,65,66] in accordance with previous studies.[6,12,20,21,74,80,82] Based on the studies' level of evidence, a moderate recommendation (grade B) was given to use massage postmatch in football, aiming to recover perceptual outcomes but not physical performance, which is in line with Abaïdia and Dupont[12]. Further research is needed to examine the potential physiological effects of massage in recovery.

Perspectives

Previous studies have rarely differentiated recovery outcomes and have interpreted recovery as a whole. This fact can create difficulties in classifying the effectiveness of a given method, as the method may be relevant in a specific dimension. The present study differentiated recovery in 3 dimensions (ie physical, physiological, and perceptual performance), as they are areas of interest with a potentially greater number of studies. In the future, other dimensions also relevant in the recovery process (eg, psychosocial) may be explored. Another perspective concerns the need to perform further investigation with better experimental designs (ie, randomized control trials), higher methodological quality (to maximize the level of evidence), and lower risk of bias, particularly for sleep, nutrition, and massage.

Finally, alternative approaches may also be sought in the recovery of physical and physiological performance parameters, since none of the methods investigated in the present had sufficient evidence to be effective. In this sense, we propose 3 paths that should drive the future of research in football postmatch recovery: (1) different exercise training approaches practiced prior to the harmful exercise could be advantageous in the recovery process, (2) combined use of recovery methods, and (3) athletes' individualization of recovery methods.

Practical Applications

Despite the limited number of studies in some methods (ie, sleep, nutrition, and massage), grades of recommendation based on methodological quality of studies included and corresponding levels of evidence were assigned to recovery of physical, physiological, and perceptive outcomes after the use of commonly used recovery methods in professional football:

- Sleep: strong recommendation against its occasional use due to its ineffectiveness to improve physical performance, physiological, and perceptive recovery;
- Nutrition: insufficient evidence to make a specific recommendation and to reject its occasional use due to its ineffectiveness to improve physical performance, physiological, and perceptive recovery;
- Cold water immersion: moderate recommendation to accept its use for perceptive recovery;
- Active recovery: moderate recommendation against its use due to its ineffectiveness to improve physical performance, physiological, and perceptive recovery;
- Massage: moderate recommendation to accept its use for perceptive recovery.

Conclusions

This study sought to gather the current evidence about the effectiveness of postmatch recovery methods in football and to provide graded recommendations for the use of methods to recover physical performance, physiological, and perceptive outcomes either in male or female athletes. Cold water immersion and massage showed effectiveness for perceptive recovery and, therefore, can be recommended to be used after football matches. Nevertheless, the application of cold water immersion is recommended in strictly necessary moments (eg, congested fixtures or after highly intense training sessions). In contrast, no recovery benefits were identified postmatch for active recovery. Additionally, the lack of randomized controlled trials, primarily attributed to the difficulty of implementing such studies in elite team sport environments, has led to a lack of evidence regarding the benefit of sleep and nutrition strategies identified in this systematic review. Overall, there is a need to increase the number of studies with higher methodological quality and level of evidence to support the use of recovery strategies in football.

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data: Querido, Radaelli, Brito, Vaz, Freitas. *Original drafting:* Querido. *Review and editing:* Querido, Radaelli, Brito, Vaz, Freitas. *Supervision:* Brito, Freitas.

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